SilverBrew

Automatic Herbal Tea Maker For Elderly

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KeyWords

Color Coded Interface

Nursing Home

Voice Support

Automatic brewing

Understandable Process

Increasing Sociality

Elderly Users

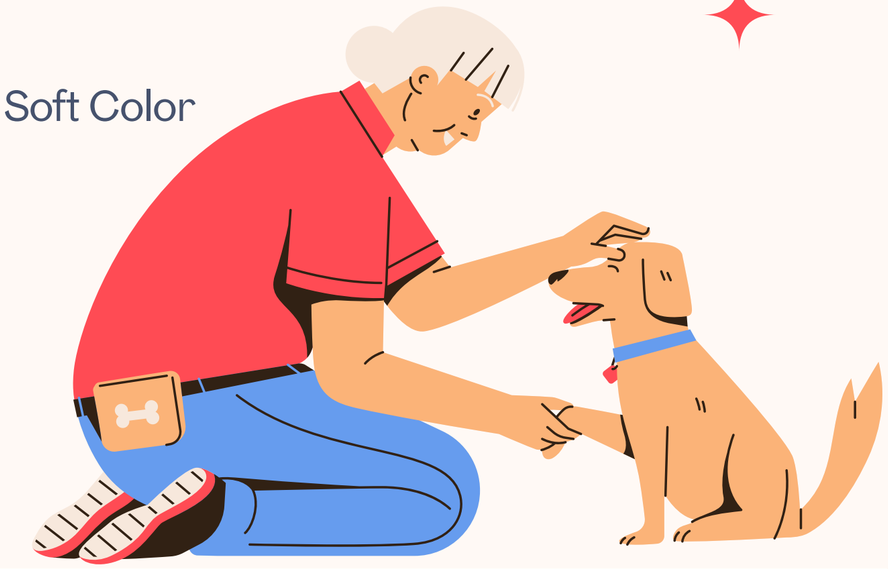
65-74
74-85

Soft Color

Easy Clean and Usage

Learned Dependency

Readability and Understandability



Key Design Milestones

- **Problem Identification:**

Research was conducted on what to pay attention to, taking into account the health problems of elderly users.
Demonstrating the potential for older adults to meet their own needs by breaking the principle of learned dependency that emerges in later life

- **Concept Development:**

Herbal tea that can be prepared with a single touch and herbal tea containers that do not require frequent refilling have been developed.

- **Interaction Design:**

High-contrast colors were used to create buttons that are easier for older adults to understand. Large, easy-to-read text was preferred, and the interface was kept as simple as possible.

- **Accessibility Improvement:**

A voice notification feature has been added, and buttons that are recognized as being pressed have been added.

- **Cleaning Mechanism Development:**

The brewing chamber and grounds cleaning system have been developed. An automatic grounds waste drawer has been designed.

My Design Decisions

- Designing the product specifically for elderly users
- Creating a simple and intuitive interface
- Using color coding for tea selection
- Adding large readable temperature display
- Implementing voice feedback for brewing completion
- Designing an automatic residue cleaning system
- Tea storage area
- Instant brewing chamber for fresh taste

Ai RoleTags

Clarifier
Artificial intelligence clarified the preferences and difficulties experienced by the elderly more clearly.

Critique
Ai critiqued what was necessary for my AI design to actually work and which mechanisms would be more useful. Beyond that, ai offered ideas on scaling my design. Ai helped me identify points in the design where older adults might struggle.

Research
The knowledge base within artificial intelligence enabled me to gather information from different sources for my research. It informed me about the specific characteristics I needed to know about the elderly and the working strategies of the mechanisms, enabling me to develop a more functional device.

Selected Impactful Prompts

- How might we enable elderly users to prepare hot beverages independently without feeling dependent on others?
- How might we reduce the risk of burns and hot water accidents while preparing hot drinks?
- How might we design a hot beverage system that elderly users can understand instantly without complex instructions?
- How can we make the cleaning process easier?
- What are main health issues that elderly have?
- Can you describe me the cleaning mechanism and give me some examples that used in other machines?
- How can I preserve the freshness of herbal teas?

Ai Responses

The AI provided design suggestions, visual renderings, and user scenario descriptions. It helped explore different interface layouts, interaction models, and mechanical solutions. AI also supported the visualization process by generating realistic product renders and storyboard illustrations.

MyReflections

Artificial intelligence really accelerated my research process, but throughout the process, I often identified problems myself through brainstorming. AI mostly agreed with me, so it couldn't provide me with sufficient criticism. It couldn't produce logical renders for mechanical issues, but it tried to explain the working principles to me with examples. I used three different AI tools for rendering, but they were somewhat unsuccessful in changing the details, and I struggled a lot when writing the prompts. Even though I used AI, my own ideas shaped the project throughout the process.

What I Accepted, Rejected, Transformed

Accepted

- Color-coded tea selection system
- Large digital display
- Automatic brewing mechanism
- Voice notification feature
- Easy clean mechanism

Rejected

- Complex multi-button interfaces
- Overly technical or complicated mechanical systems
- Excessive visual information on the interface
- Unsafe water spill area

Tranformed

- Cleaning mechanism was simplified into a rear waste drawer system
- Interface icons were redesigned to be simpler and more readable
- Render style was adjusted to highlight the product clearly.
- Volume control buttons
- Waste drawer